Faster NTRU-based Bootstrapping with Extended and Sorting-based Techniques

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Abstract. NTRU-based bootstrapping offers a high-performance variant of FHEW-like bootstrapping schemes and is simpler than its RLWE-based counterparts. Nevertheless, it remains costly in terms of both memory and time. A key computational bottleneck arises from the constraint $q \mid 2N$ which links the LWE ciphertext modulus q to the NTRU dimension N. In this work, we apply extended techniques to remove the limitation $q \mid 2N$. With a large modulus q, we are able to complete the bootstrapping algorithms using a relatively small N-dimensional ring, thereby improving both time and memory efficiency. Additionally, we employ sorting-based techniques to eliminate unnecessary operations, which further enhances time performance. We also extend this approach to improve the efficiency of NTRU-based multi-key bootstrapping algorithms.

We implement our algorithms using the CHIFHE and MKFHE libraries. At the 128-bit security level, our sorting-based techniques (with factor $\eta = 16$) accelerate single-key bootstrapping by factors of 1.82× and 1.24× over the FINAL scheme (ASIACRYPT 2021) and the key unrolling method (CHES 2024), respectively. For multi-key bootstrapping (with factor $\eta = 4$), we achieve a 1.70× speedup over the prior art (ASIACRYPT 2024). In addition, both single-key and multi-key algorithms reduce memory usage by a factor of η.

Keywords: Fully Homomorphic Encryption · Bootstrapping · NTRU · Extended (Sorting-based) Techniques

1 Introduction

Fully Homomorphic Encryption (FHE) is a family of encryption schemes that enables computations to be performed directly on ciphertexts without decryption. Owing to this powerful property, FHE has found numerous applications in privacy-preserving computation, including privacy-preserving machine learning (PPML) [CDPP22] and private information retrieval (PIR) [HHCG+23]. Existing efficient FHE constructions, such as BGV [BGV12], B/FV [Bra12, FV12], FHEW/TFHE [DM15, CGGI16] and CKKS [CKKS17], are typically based on the hardness of the LWE/RLWE [Reg09, LPR10] or NTRU [HPS98] problems. These schemes introduce noise into ciphertexts, and the noise grows during homomorphic computations, potentially causing decryption failures once it exceeds a certain threshold. Bootstrapping [Gen09] was introduced to refresh ciphertexts and reduce the accumulated noise to a manageable level. Since bootstrapping requires the homomorphic evaluation of the decryption function, it is typically the most computationally expensive procedure in an FHE scheme.

In this paper, we focus on NTRU-based FHE schemes [BIP+22, XZD+23, LLW+24], which belong to the family of FHEW/TFHE-style schemes. The original FHEW schemes [ASP14, DM15, LMK+23] and TFHE schemes [GINX16, CGGI16, CGGI17] employ RLWE ciphertexts to re-encrypt the input LWE ciphertext during bootstrapping. In 2022, Bonte et al. [BIP+22] proposed replacing RLWE ciphertexts with NTRU ciphertexts to perform the re-encryption and homomorphic decryption procedures. This approach significantly improves the efficiency of bootstrapping because an NTRU ciphertext consists of only one polynomial, whereas an RLWE ciphertext typically consists of two polynomials. Subsequently, Xiang et al. [XZD+23] employed automorphisms to implement blind rotation for Gaussian-distributed LWE secret keys over $\mathbb{Z}_q^n$. Li et al [LLW+24] used the key unrolling [BMMP18] and the generator method [LMK+23] to improve the efficiency of blind rotation with (LWE) secret key distribution of uniform over $\{0,1\}^n$ and Gaussian over $\mathbb{Z}_q^n$, respectively.

Although these works have significantly improved the efficiency of NTRU-based bootstrapping, the blind rotation step in bootstrapping remains computationally expensive. One important reason is that the NTRU ring dimension N is required to satisfy the condition $q \mid 2N$, where q denotes the ciphertext modulus. In some applications, q can be very large; for example, q may be as large as 2^{32} when evaluating Boolean functions [BPR24] or in PPML and PIR applications [MW24, CDPP22]. Under the constraint $q \mid 2N$, a large ciphertext modulus inevitably leads to a large NTRU dimension N. Since blind rotation, the dominant computational bottleneck in bootstrapping, involves numerous operations over an N-dimensional polynomial ring, a large value of N can substantially degrade the efficiency of bootstrapping. This naturally raises the following question:

Can we choose the NTRU dimension N such that $2N < q$ when the ciphertext modulus q becomes large?

Traditional (Single-key) FHE schemes generally support homomorphic computations only on ciphertexts encrypted under the same secret key, making them unsuitable for multi-user scenarios in which individual data owners require independent key ownership. To address this limitation, Multi-Key Homomorphic Encryption (MK-FHE) [LATV12] was introduced in 2012. MK-FHE enables joint homomorphic computations over ciphertexts encrypted under distinct secret keys. However, bootstrapping for MK-FHE [CCS19, KMS24, XZW+25] remains computationally expensive. In this paper, we also investigate NTRU-based bootstrapping for MK-FHE. Existing work [XZW+25] similarly requires the relation $q \mid 2N$ between the ciphertext modulus q and the NTRU dimension N, which may become a significant efficiency bottleneck as the modulus q increases.

1.1 Our Contributions and Technical Overview

In this paper, we improve the efficiency of NTRU-based bootstrapping primarily by eliminating the requirement that $q \mid 2N$ where q is the LWE ciphertext modulus and N is the NTRU dimension. This relaxation leads to improvements in both memory consumption and computational efficiency. We further generalize our techniques to multi-key NTRU-based bootstrapping. Our main technical contributions are summarized below.

First, we employ the extended bootstrapping technique (EBS) [LY23] to reduce both the memory size and the computational cost of single-key bootstrapping. Specifically, let $R_{Q,N} = \mathbb{Z}_Q[X]/\langle X^N + 1\rangle$. We choose N_{ext} such that $q \mid 2N_{\text{ext}}$, and set $N = N_{\text{ext}}/\eta$ so that the problem $\text{NTRU}_{N,Q,\chi}$ is hard. Then we have an isomorphism $\tau : R_{Q,N_{\text{ext}}} \to R_{Q,N}^{\eta}$. The bootstrapping algorithm takes the polynomial $T(X) \cdot X^{-\hat{b}}$ as input, and applies τ to obtain η polynomials $\{\tau_j(T(X) \cdot X^{-\hat{b}})\}_{j\in[\eta]}$. Then each polynomial takes one external product with the initial bootstrapping key $\mathbf{evk}'$ to get η (N-dim) NTRU ciphertexts $\{\text{ACC}_j \in \text{NTRU}_{f,Q}^N(\tau_j(T(X) \cdot X^{-\hat{b}})\}_{j\in[\eta]}$. During blind rotation, we perform $\eta \cdot (n-1)$

times N-dimensional external products to obtain

$$\{\mathrm{ACC}_j \in \mathrm{NTRU}_{f,Q}^{N}(\tau_j(T(X)\cdot X^{-\hat{b}+\sum_{i=0}^{n-2} a_i\cdot s_i}))\}_{j\in[\eta]}.$$

One additional N-dimensional external product is then performed to obtain

$$\mathrm{ACC} \in \mathrm{NTRU}_{f,Q}^{N}(\tau_0(T(X)\cdot X^{-\hat{b}+\sum_{i=0}^{n-1} a_i\cdot s_i})).$$

Consequently, our blind-rotation procedure requires $\eta\cdot(n-1)+1$ dimension N external products, whereas previous approaches require n dimension $(N\cdot\eta)$ external products. Moreover, the bootstrapping and key-switching keys require $(n+1)\cdot d\cdot N\cdot\log_2 Q$ bits and $2\cdot N\cdot B_{ks}\cdot d_{ks}\cdot\log_2 Q_{ks}$ bits of storage, respectively. Since the working dimension is reduced from N_{ext} to N, the corresponding memory consumption is reduced by a factor of η.

Next, we apply the sorting-based bootstrapping technique (SBS) [BORLT26] to further improve computational efficiency. We observe that, in order to compute

$$\mathrm{ACC} \in \mathrm{NTRU}_{f,Q}^{N}(\tau_0(T(X)\cdot X^{-\hat{b}+\sum_{i=0}^{n-1} a_i\cdot s_i})),$$

only two ciphertexts from the previous step are needed. By appropriately rearranging the order of the external products, we can therefore eliminate a number of unnecessary operations. As a result, the proposed method requires only $n\cdot\left(\frac{2\eta^2+1}{3\eta}\right)$ (N-dim) external products (in average).

Finally, we extend the EBS and SBS techniques to NTRU-based multi-key fully homomorphic encryption (MK-FHE) [XZW+25]. As in the single-key setting, our techniques improve the efficiency of the hybrid product, which is the dominant and most computationally expensive operation in multi-key bootstrapping. Consequently, the proposed multi-key bootstrapping scheme achieves both lower computational cost and reduced memory consumption.

1.2 Related Work

For FHEW/TFHE bootstrapping, Bernard and Joye [BJ26] proposed automorphism-parametrized external product to decrease the times of key switch operations in the blind rotation, leading to significant improvement of bootstrapping algorithm with Gaussian secret keys. Hwang et al [HMS26] used the slot-encoding instead of coefficient-encoding. By using this method, the blind rotation can cover the entire plaintext sapce.

For MK-FHE, Kwak et al [KMS24] proposed a new framework for the Hybrid Product operation. Xiang et al [XMH+26] presented a new bootstrapping algorithm by using packing method and HomTrace function. They reduce the times of external product from $O(dkn)$ to $O(kn)$, and further to $O(\sqrt{kn})$ in parallel.

1.3 Paper Organization

In Section 2, we review the necessary background on NTRU-based bootstrapping. Section 3 presents our optimized NTRU-based bootstrapping algorithms. In Section 4, we extend these techniques to improve the efficiency of multi-key bootstrapping. Section 5 provides a comparison of the optimized algorithms, together with parameter settings and implementation results. Finally, we conclude this paper in Section 6.

2 Preliminaries

2.1 Notations

The bold letters (e.g. $\mathbf{a}$) represent vectors. For the inner product between two vectors $\mathbf{a}, \mathbf{b}$, we write $\langle \mathbf{a}, \mathbf{b} \rangle$. Let $\|\cdot\|_2$ and $\|\cdot\|_\infty$ denote the 2-norm and ∞-norm for a vector, respectively. We use $\lfloor \cdot \rfloor$, $\lceil \cdot \rceil$, $\lfloor \cdot \rceil$ to denote the floor, ceiling and round function on float numbers, respectively. Let $\mathbb{Z}$ be the ring of integers. For $q \in \mathbb{Z}$, denote $\mathbb{Z}_q = \mathbb{Z}/q\mathbb{Z}$ to be the residue ring modulo q, and the notation $[a]_q$ means $a \pmod q$ for $a \in \mathbb{Z}$. We use $[n]$ to represent the set $\{0, 1, ..., n-1\}$ for some positive integer n. The arrow $\leftarrow$ is used to represent randomly chosen an element from a distribution. Denote the variance of a random variable a over $\mathbb{Z}_q$ by $\mathrm{Var}(a)$. The error distribution χ used in this paper refers to the discrete Gaussian distribution over $\mathbb{Z}_q$ or $\mathbb{Z}_q^n$. A look up table (LUT) [CGGI17] means a map $h : \mathbb{Z}_t \to \mathbb{Z}_t$.

Let N be a power of 2. We use R_N to denote the $2N$-th cyclotomic ring $\mathbb{Z}[X]/\langle X^N + 1 \rangle$. For a positive integer q, let $R_{q,N}$ be the quotient ring R_N/qR_N. For $a \in R_N$, $\phi(a) = (a_0, a_1, ..., a_{N-1})$ is the coefficient vector of a, and the variance of a is defined as $\mathrm{Var}(a) = \mathrm{Var}(\phi(a))$. The norm of a is defined as $\|a\|_2 = \|\phi(a)\|_2$ and $\|a\|_\infty = \|\phi(a)\|_\infty$. The notation $U(S)$ represents the uniform random distribution for a set S.

2.2 Gadget Decomposition

We describe a technique known as gadget decomposition [CGGI16], which is commonly used to reduce noise in the ciphertexts.

Let Q be the ciphertext modulus and B be the decomposition base. We compute $d = \lceil \log_B Q \rceil$ and define the gadget vector $\mathbf{g} = (1, B, ..., B^{d-1})$. For a polynomial $a \in R_{Q,N}$, we define the signed decomposition with the gadget vector $\mathbf{g}$ as

$$\mathbf{g}^{-1}(a) = \left([a]_B, \left[\left\lfloor \frac{a}{B} \right\rceil \right]_B, \cdots, \left[\left\lfloor \frac{a}{B^{d-1}} \right\rceil \right]_B \right) = (a_0, ..., a_{d-1}).$$

We have $\|a_i\|_\infty \leqslant B/2$ for $i \in [d]$ and $\left[\langle \mathbf{g}^{-1}(a), \mathbf{g} \rangle \right]_Q = a$.

2.3 Hard Problems and Ciphertexts

We recall some hard problems which will be used in this paper.

Definition 1 (Decisional LWE Problem [Reg09])**.** *Suppose q and n are positive integers, and χ is the discrete Gaussian distribution over $\mathbb{Z}_q$. The decisional LWE problem* $\mathrm{LWE}_{n,q,\chi}$ *is to distinguish between the distribution $(\mathbf{a}, \langle \mathbf{a}, \mathbf{s} \rangle + e) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ and the uniform distribution over $\mathbb{Z}_q^{n+1}$, where $\mathbf{a} \in \mathbb{Z}_q^n$ is chosen uniformly at random, $e \leftarrow \chi$ and $\mathbf{s}$ is the secret key that is chosen from $\mathbb{Z}^n$.*

Definition 2 (Decisional NTRU Problem [HPS98])**.** *Suppose Q and N are positive integers, χ is an error distribution over $R_{Q,N}$. The decisional NTRU problem* $\mathrm{NTRU}_{N,Q,\chi}$ *is to distinguish between the distribution $g_e \cdot f^{-1} \in R_{Q,N}$ and the uniform distribution over $R_{Q,N}$, where $g_e \leftarrow \chi$ is the noise polynomial, and the secret key f is invertible in $R_{Q,N}$.*

Definition 3 (Decisional Vectorized-NTRU Problem [XZD+23])**.** *Suppose Q and N are positive integers, and χ is an error distribution over $R_{Q,N}$. For a positive integer d, the decisional vectorized-NTRU problem* $\mathrm{VNTRU}_{N,Q,\chi}$ *is to distinguish between the distribution $\mathbf{g} \cdot f^{-1} \in R_{Q,N}^d$ and the uniform distribution over $R_{Q,N}^d$, where $\mathbf{g}_e \leftarrow \chi^d$ is the noise polynomial vector, and the secret key f is invertible in $R_{Q,N}$.*

For LWE ciphertexts, we use the B/FV-like ciphertexts [Bra12, FV12]:

Definition 4 (LWE Ciphertexts)**.** *Suppose q and n are positive integers, t is the modulus of the message space with $t \mid q$, and χ is an error distribution over $\mathbb{Z}_q$. The LWE ciphertext of a message $m \in \mathbb{Z}_t$ under the (binary) secret key $\mathbf{s} \in \mathbb{Z}_q^n$ is defined as*

$$\mathrm{LWE}_{\mathbf{s},q}^n(m) = \{(\mathbf{a}, b) \in \mathbb{Z}_q^{n+1} \mid b = \left[\langle \mathbf{a}, \mathbf{s}\rangle + \left\lfloor \frac{q}{t} \right\rceil \cdot m + e\right]_q, \mathbf{a} \leftarrow U(\mathbb{Z}_q^n), e \leftarrow \chi_e\}.$$

Definition 5 (Multi-Key LWE Ciphertexts)**.** *Suppose q and n are positive integers, t is the modulus of the message space with $t \mid q$, and χ is an error distribution over $\mathbb{Z}_q$. The Multi-key LWE ciphertext (with k parties) of a message $m \in \mathbb{Z}_t$ under k secret keys $\{\mathbf{z}_i \in \mathbb{Z}_q^n\}_{1 \leqslant i \leqslant k}$ is defined as*

$$\mathrm{M.LWE}_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}^n(m) = \{(b, \mathbf{a}_1, ..., \mathbf{a}_k) \in \mathbb{Z}_q^{k\cdot n+1} \mid b = \left[\sum_{i=1}^{k} \langle \mathbf{a}_i, \mathbf{z}_i\rangle + \left\lfloor \frac{q}{t} \right\rceil \cdot m + e\right]_q,$$
$$\mathbf{a}_i \leftarrow U(\mathbb{Z}_q^n), e \leftarrow \chi_e\}.$$

Remark. In this paper, LWE secret keys are chosen from binary distribution, i.e., $\mathbf{s}, \mathbf{z}_i \in \{0,1\}^n$.

Next, we introduce the NTRU ciphertexts used in bootstrapping.

Definition 6 (NTRU Ciphertexts)**.** *Let Q be a positive integer. Define the (NTRU) secret key distribution χ_f as*

$$\chi_f = \{f = \sum_{i=0}^{N-1} f_i \cdot X^i \in R_N \mid f_i \in \{-1, 0, 1\},\ f \text{ is invertible in } R_{Q,N}.\}$$

The N-dimensional (scalar) NTRU ciphertext of a message polynomial $m \in R_Q$ under the secret key $f \in \chi_f$ is defined as

$$\mathrm{NTRU}_{f,Q}^N(m) = \{c \in R_{Q,N} \mid c = \left[(g_e + m) \cdot f^{-1}\right]_Q, g_e \leftarrow \chi_e'\},$$

where the distribution χ_e' is an error distribution over $R_{Q,N}$.

The N-dimensional vector-NTRU ciphertext (under the secret key $f \in \chi_f$) is defined as

$$\mathrm{VNTRU}_{f,Q}^N(m) = \{\mathbf{c} \in R_{Q,N}^d \mid \mathbf{c} = \left[\mathbf{g}_e \cdot f^{-1} + \mathbf{g} \cdot m\right]_Q,\ \mathbf{g}_e \leftarrow {\chi_e'}^d\},$$

where the distribution ${\chi_e'}^d$ is a d-dimensional error distribution with $d = \lceil \log_B Q \rceil$, and the gadget vector $\mathbf{g}$ is defined as in section 2.2 .

Definition 7 (Multi-Key NTRU Ciphertexts)**.** *Let Q be a positive integer. The Multi-key N-dimensional (scalar) NTRU ciphertext (with k parties) of a message polynomial $m \in R_Q$ under the secret key $\{s_i \in \chi_f\}_{1 \leqslant i \leqslant k}$ is defined as*

$$\mathrm{M.\,NTRU}_{\{s_i\}_{1\leqslant i\leqslant k},Q}^N(m) = \{(c_1, ..., c_k) \in R_{Q,N}^k \mid \sum_{i=1}^{k} [c_i \cdot s_i]_Q = g_e + m, g_e \leftarrow \chi_e'\},$$

where the distribution χ_e' is an error distribution over $R_{Q,N}$.

Lemma 1 (External Product [XZD+23])**.** *Let $c = \mathrm{NTRU}_{f,Q}^N(u) \in R_{Q,N}$ be an NTRU ciphertext with noise variance $\mathrm{Var}(c)$. Let $\mathbf{c}' = \mathrm{VNTRU}_{f,Q}^N(v)$ be a vector-NTRU ciphertext with noise variance $\mathrm{Var}(\mathbf{c}')$. Define the N-dimensional external product $\odot_N$ as*

$$c \odot_N \mathbf{c}' = \left\langle \mathbf{g}^{-1}(c),\ \mathbf{c}' \right\rangle := \hat{c}.$$

Then we have $\hat{c} = (\hat{g}_e + u \cdot v) \cdot f^{-1}$ is an NTRU ciphertext for $u \cdot v$ (i.e., $\hat{c} = \mathrm{NTRU}_{f,Q}^N(u \cdot v)$), whose variance satisfies

$$\mathrm{Var}(\hat{c}) = \mathrm{Var}(\hat{g}_e) \leqslant d \cdot N \cdot \frac{B^2}{12} \cdot \mathrm{Var}(\mathbf{c}') + \|v\|_2^2 \cdot \mathrm{Var}(c).$$

2.4 Bootstrapping

We firstly recall several results that play key roles in the bootstrapping procedure.

Lemma 2 (Sample Extraction [LLW+24, XZD+23]). *Suppose we have an NTRU ciphertext*

$$c \in \mathrm{NTRU}^N_{f,Q}(m), \text{ where } c = \sum_{i=0}^{N-1} c_i \cdot X^i, \ m = \sum_{i=0}^{N-1} m_i \cdot X^i.$$

Then the sample extraction algorithm outputs

$$\mathrm{SampleExtract}(c) = (0, c_0, -c_{N-1}, -c_{N-2}, \ldots, -c_1) \in \mathbb{Z}_Q^{N+1},$$

which satifies $\mathrm{SampleExtract}(c) \in \mathrm{LWE}^N_{\phi(f),Q}(m_0)$.

Lemma 3 (Key Switching [BGV12, Bra12]). *Let Q_{ks}, B_{ks} be two positive integers and $d_{ks} = \lceil \log_{B_{ks}} Q_{ks} \rceil$. Suppose we have an LWE ciphertext* $\mathrm{ct} = (b, \mathbf{a}) \in \mathrm{LWE}^N_{\phi(f),Q_{ks}}(m)$ *with noise variance* $\mathrm{Var}(\mathrm{ct})$*, and the switching keys* $\mathbf{ksk}_{\phi(f)\to\mathbf{s}} = \{\mathrm{ksk}_{i,j,v} \in \mathrm{LWE}^n_{\mathbf{s},Q_{ks}}(v f_i B_{ks}^j)\}$*, where $v \in [B_{ks}]$, $i \in [N]$, $j \in [d_{ks}]$, with noise variance* $\mathrm{Var}(\mathbf{ksk})$*. Suppose* $\mathbf{a} = (a_0, ..., a_{N-1})$*, a_i has gatget decomposition* $\mathbf{g}_k^{-1}(a_i) = (a_{i,0}, ..., a_{i,d_{ks}-1})$ *for $i \in [N]$ with gadget vector* $\mathbf{g}_k = (1, B_{ks}, ..., B_{ks}^{d_{ks}-1})$*. Then the LWE key switching algorithm outputs*

$$\mathrm{KeySwitch}_{\phi(f)\to\mathbf{s}}(\mathrm{ct}) = (b, 0) - \sum_{i=0}^{N-1} \sum_{j=0}^{d_{ks}-1} \mathrm{ksk}_{i,j,a_{i,j}} \in \mathbb{Z}_q^{n+1},$$

which satifies $\mathrm{ct}' = \mathrm{KeySwitch}_{\phi(f)\to\mathbf{s}}(\mathrm{ct}) \in \mathrm{LWE}^n_{\mathbf{s},Q_{ks}}(m)$*. We also have the noise variance*

$$\mathrm{Var}(\mathrm{ct}') \leqslant \mathrm{Var}(\mathrm{ct}) + N \cdot d_{ks} \cdot \mathrm{Var}(\mathbf{ksk}).$$

Lemma 4 (Modulus Switching [BGV12, Bra12]). *Let* $\mathrm{ct} = (b, \mathbf{a}) \in \mathrm{LWE}^n_{\mathbf{s},Q}(m)$ *be an LWE ciphertext with error variance* $\mathrm{Var}(\mathrm{ct})$*. Then the modulus switching algorithm outputs*

$$\mathrm{ModSwitch}_{Q\to q}(\mathrm{ct}) = \left(\left\lfloor \frac{q}{Q} \cdot b \right\rceil, \ \left\lfloor \frac{q}{Q} \cdot \mathbf{a} \right\rceil \right) \in \mathbb{Z}_q^{n+1},$$

which satifies $\mathrm{ct}' = \mathrm{ModSwitch}_{Q\to q}(\mathrm{ct}) \in \mathrm{LWE}^n_{\mathbf{s},q}(m)$*. We also have the noise variance*

$$\mathrm{Var}(\mathrm{ct}') \leqslant \frac{q^2}{Q^2} \cdot \mathrm{Var}(\mathrm{ct}) + \frac{n \cdot \|\mathbf{s}\|_\infty + 1}{12}.$$

The purpose of bootstrapping is to reduce the noise in a ciphertext to the level of a freshly encrypted ciphertext. Chilotti et al. [CGGI17] introduced functional bootstrapping, which enables the evaluation of a lookup table (LUT) on the plaintext during the bootstrapping procedure. The functional bootstrapping algorithm consists of five main steps: initialization, blind rotation, sample extraction, key switching, and modulus switching.

Initialization. For an input LWE ciphertext $(b, \mathbf{a}) \in \mathrm{LWE}^n_{\mathbf{s},q}(m)$, an LUT h, and a special bootstrapping key, compute the test polynomial $T(X)$ and output an NTRU ciphertext $c \in \mathrm{NTRU}^N_{f,Q}(T(X) \cdot X^{-\frac{2N}{q} \cdot b})$;

Blind Rotation. Input the first member $\mathbf{a}$ of the LWE ciphertext $(b, \mathbf{a})$, the initial NTRU ciphertext $\mathrm{NTRU}^N_{f,Q}(T(X) \cdot X^{-\frac{2N}{q} \cdot b})$ and the bootstrapping keys. Use external products to homomorphically multiply $X^{\frac{2N}{q} \cdot \sum_{i=0}^{n-1} a_i s_i}$ and output

$$c \in \mathrm{NTRU}^N_{f,Q}\left(T(X) \cdot X^{\frac{2N}{q} \cdot (-b + \sum_{i=0}^{n-1} a_i s_i)} \right);$$

Sample Extraction. Input the NTRU ciphertext

$$c \in \mathrm{NTRU}_{f,Q}^{N}\left(T(X)\cdot X^{\frac{2N}{q}\cdot(-b+\sum_{i=0}^{n-1}a_i s_i)}\right),$$

output an LWE ciphertext $(b,\mathbf{a}) \in \mathrm{LWE}_{\phi(f),Q}^{N}(h(m))$ as in Lemma 2;
Key Switch. Input $(b,\mathbf{a}) \in \mathrm{LWE}_{\phi(f),Q}^{N}(m)$ and output $(\mathbf{a}',b') \in \mathrm{LWE}_{\mathbf{s},Q}^{n}(h(m))$ as in Lemma 3;
Modulus Switch. Input $(\mathbf{a}',b') \in \mathrm{LWE}_{\mathbf{s},Q}^{n}(m)$ and output $(\mathbf{a}',b') \in \mathrm{LWE}_{\mathbf{s},q}^{n}(h(m))$ as in Lemma 4.

3 Single-Key Bootstrapping with Extended and Sorting-based Techniques

In this section, we present our single-key NTRU-based bootstrapping scheme incorporating the extended and sorting-based techniques. We begin by establishing an important result, which we refer to as the Extended Isomorphism.

3.1 Extended Isomorphism

Lemma 5 ([BORLT26], [LY23]). *Let $\eta \in \mathbb{Z}$ such that $\eta \mid N$. Let ι be the map*

$$\iota : R_{q,N/\eta} \longrightarrow R_{q,N},\quad f(X)=\sum_{i=0}^{N/\eta-1} f_i\cdot X^i \longmapsto \sum_{i=0}^{N/\eta-1} f_i\cdot X^{\eta i}.$$

Then, ι is a ring homomorphism.

We note that $R_{q,N}$ can be seen as a $R_{q,N/\eta}$-module. The scalar multiplication $f(X)\bullet m(X)$ between $f(X)\in R_{q,N/\eta}$ and $m(X)\in R_{q,N}$ can be defined as:

$$f(X)\bullet m(X)=\iota(f(X))\cdot m(X)\in R_{q,N}.$$

Now we present the extended isomorphism:

Lemma 6. *Let τ be a map defined as*

$$\tau : R_{q,N} \longrightarrow R_{q,N/\eta}^{\eta},\quad m(X)=\sum_{i=0}^{N-1} m_i X^i \longmapsto (m_0(X),m_1(X),...,m_{\eta-1}(X)),$$

where $m_j(X)=\sum_{i=0}^{N/\eta-1} m_{i\eta+j}X^i$ for $j=0,1,...,\eta-1$. Then τ is a $R_{q,N/\eta}$-module isomorphism.

Proof. First we prove τ is invertible. Consider the map $\hat{\tau}$:

$$\hat{\tau} : R_{q,N/\eta}^{\eta} \to R_{q,N}$$

$$(m_0(X),m_1(X),...,m_{\eta-1}(X)) \longmapsto m(X)=\sum_{i=0}^{N-1} m_i X^i, m_j(X)=\sum_{i=0}^{N/\eta-1} f_{i\eta+j}X^i.$$

Then we have

$$\hat{\tau}(\tau(m(X)))=m(X),\ m(X)\in R_{q,N},$$

$$\tau(\hat{\tau}\,(m_0(X),m_1(X),...,m_{\eta-1}(X)))=(m_0(X),m_1(X),...,m_{\eta-1}(X)),(m_i(X))_{i\in[\eta]}\in R_{q,N/\eta}^{\eta}.$$

Besides, τ is a module-homomorphism since:

$$\tau(m_1(X)+m_2(X)) = \tau(m_1(X)+m_2(X)), m_1, m_2 \in R_{q,N},$$
$$\tau(m(X) \bullet \iota(f(X)) = f(X) \cdot \tau(m(X)), m \in R_{q,N}, f \in R_{q,N/\eta}.$$

□

We also define the map

$$\tau_j : R_{q,N} \longrightarrow R_{q,N/\eta}, \quad m(X) \longmapsto m_j(X).$$

where $m_j(X)$ is the j-th component of $\tau(m)$. Before describing our bootstrapping, we show how to compute $\tau(m(X) \cdot X^a)$.

Lemma 7 ([BORLT26])**.** *Let τ be the function defined in Lemma 6. Let $m(X) = \sum_{i=0}^{N-1} m_i X^i$ be a polynomial in $R_{q,N}$ such that $\tau(m(X)) = (m_0(X), \cdots, m_{\eta-1}(X))$ with $m_i(X) = \sum_{j=0}^{N-1} m_{j\eta+i} X^j$ a polynomial in $R_{q,N}$ for $i \in [\eta]$.*

For any $a \in \mathbb{Z}$ we have $\tau_j(m(X) \cdot X^a) = m_{[(j-a)]_\eta}(X) \cdot X^{\lceil \frac{a-j}{\eta} \rceil}$.

Proof. See Lemma 5 of [BORLT26].

Corollary 1. *Let a be a positive integer. If $c \in \mathrm{NTRU}^N_{f,Q}(\tau_{[j-a]_\eta}(m))$, then*

$$c \cdot X^{\lceil \frac{a-j}{\eta} \rceil} \in \mathrm{NTRU}^N_{f,Q}(\tau_j(m \cdot X^a)).$$

Moreover, in the multi-key NTRU ciphertext, if $\mathbf{c} \in \mathrm{M.NTRU}^N_{\{s_i\}_{1 \leqslant i \leqslant k},Q}(\tau_{[j-a]_\eta}(m))$, we have

$$\mathbf{c} \cdot X^{\lceil \frac{a-j}{\eta} \rceil} \in \mathrm{M.NTRU}^N_{\{s_i\}_{1 \leqslant i \leqslant k},Q}(\tau_j(m \cdot X^a)).$$

3.2 Our Bootstrapping with Extended Techniques

We now describe our NTRU-based bootstrapping scheme using the extended techniques. Let q be the modulus of LWE ciphertexts. We first choose N_{ext} such that $q \mid 2N_{\text{ext}}$ and $N \mid N_{\text{ext}}$ while ensuring that $\mathrm{NTRU}_{N,Q,\chi}$ remains hard. Recall the SampleExtract step requires an NTRU encryption of a polynomial message m, whose contstant term m_0 is equal to the constant coefficient of

$$T(X) \cdot X^{-\hat{b}+\sum_{i=0}^{n-1} \hat{a}_i \cdot s_i}.$$

It is straightforward to observe that the constant coefficient of

$$\tau_0(T(X) \cdot X^{-\hat{b}+\sum_{i=0}^{n-1} \hat{a}_i \cdot s_i})$$

satisfies this property. satisfies this requirement. Following [BIP+22, LLW+24], we use the operation defined below, referred to as the CMux gate, to obtain the desired NTRU ciphertext.

Lemma 8 (Cmux Gate)**.** *Let a be a positive integer, and $\mathbf{evk} \in \mathrm{VNTRU}^N_{f,Q}(s)$ for $s \in \{0,1\}$. If there are η ciphertexts*

$$\{\mathrm{ACC}_j \in \mathrm{NTRU}^N_{f,Q}(\tau_j(m))\}_{j \in [\eta]},$$

then we compute

$$\overline{\mathrm{ACC}}_j = \mathrm{ACC}_j + (\mathrm{ACC}_{[j-a]_\eta} \cdot X^{\lceil \frac{a-j}{\eta} \rceil} - \mathrm{ACC}_j) \odot_N \mathbf{evk} \text{ for } j \in [\eta].$$

The output ciphertexts satisfies

$$\overline{\mathrm{ACC}}_j \in \mathrm{NTRU}^N_{f,Q}(\tau_j(m \cdot X^{a \cdot s})).$$

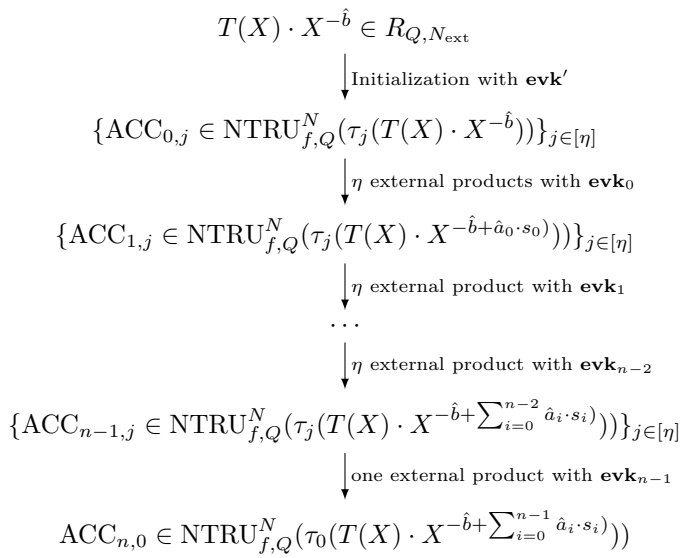


Figure 1: Our Blind Rotation for NTRU-based Bootstrapping

Proof. This can be directly derived from Lemma 1 and Corollary 1 □

Figure 1 illustrates the blind-rotation procedure of our scheme. We begin with the polynomial $T(X)\cdot X^{-\hat{b}}\in R_{Q,N_{\text{ext}}}$ and apply the isomorphism τ to obtain $\eta = N_{\text{ext}}/N$ polynomials $\{\tau_j(T(X)\cdot X^{-\hat{b}})\}_{j\in[\eta]}$. We then perform $n-1$ iterations, each of which involves η Cmux gate, After the $(n-1)$-th iteration, we get η ciphertexts

$$\{\text{ACC}_{n-1,j}\in \text{NTRU}^N_{f,Q}(\tau_j(T(X)\cdot X^{-\hat{b}+\sum_{i=0}^{n-2}\hat{a}_i\cdot s_i)}))\}_{j\in[\eta]}.$$

Using Lemma 8, we can compute $\text{ACC}_{n,0}\in\tau_0(T(X)\cdot X^{-\hat{b}+\sum_{i=0}^{n-1}\hat{a}_i\cdot s_i})$ by executing a Cmux gate :

$$\text{ACC}_{n,0}=\text{ACC}_{n-1,0}+(\text{ACC}_{n-1,[-\hat{a}_{n-1}]_\eta}\cdot X^{\left\lceil\frac{\hat{a}_{n-1}}{\eta}\right\rceil}-\text{ACC}_{n-1,0})\odot_N \mathbf{evk}_{n-1},$$

where $\mathbf{evk}_{n-1}\in\text{VNTRU}^N_{f,Q}(s_{n-1})$.

The complete bootstrapping procedure is presented in Algorithm 1. Compared with previous approaches, our method evaluate $(\eta\cdot n+1)$ dimension-N external products instead of $(n+1)$ external products of dimension-$(N\cdot\eta)$. Moreover, the dimension of the input ciphertext to the KeySwitch procedure is reduced from $N\cdot\eta$ dimension to N dimension. Now we establish the correctness of the proposed algorithm and analyze its noise variance.

Theorem 1 (Extended Bootstrapping)**.** *Let $h:\mathbb{Z}_t\to\mathbb{Z}_t$ be an LUT,* $\text{ct}\in\text{LWE}^n_{\boldsymbol{s},q}(m)$ *be an LWE ciphertext under a secret key $\boldsymbol{s}\leftarrow\{0,1\}^n$, and $\mathbf{evk}',\{\mathbf{evk_i}\}_{i\in[n]},\mathbf{ksk}_{\phi(f)\to\mathbf{s}}$ are described as in Algorithm 1. Then Algorithm 1 outputs an LWE ciphertext* $\text{ct}'\in\text{LWE}^n_{\boldsymbol{s},q}(h(m))$ *with reduced noise. Its noise variance satisfies*

$$\text{Var}(\text{ct}')\leqslant\frac{q^2}{Q_{ks}^2}\cdot\left(\frac{Q_{ks}^2}{Q^2}\cdot\sigma^2_{\text{BR}}+\sigma^2_{\text{MS1}}+\sigma^2_{\text{KS}}\right)+\sigma^2_{\text{MS2}},$$

where $\sigma^2_{\text{BR}}=(n+1)\cdot d\cdot N\cdot\frac{B^2}{12}\cdot\sigma^2$, $\sigma^2_{\text{MS1}}=\frac{N+1}{12}$, $\sigma^2_{\text{KS}}=N\cdot d_{ks}\cdot\sigma_L^2$, $\sigma^2_{\text{MS2}}=\frac{n+1}{12}$. Here, σ^2 denotes the noise variance of NTRU evaluation keys and σ_L^2 denotes the noise variance of LWE keyswitch keys.

Algorithm 1 NTRU-based Bootstrapping with Extended Techniques

Input: An LWE sample $\mathrm{ct} = (b, \mathbf{a}) \in \mathrm{LWE}^n_{\mathbf{s},q}(m)$.
The evaluation keys $\{\mathbf{evk}_i\}, i \in [n], \mathbf{evk}_i \in \mathrm{VNTRU}^N_{f,Q}(s_i)$.
A special evaluation key $\mathbf{evk}' = \mathrm{VNTRU}^N_{f,Q}(f^{-1})$.
An LWE key switching key $\mathbf{ksk}_{\phi(f)\to\mathbf{s}}$ described in Lemma 3.
Output: An LWE sample $\mathrm{ct}' \in \mathrm{LWE}^n_{\mathbf{s},q}(h(m))$.
Set $T(X) = \sum_{i=0}^{N_{\mathrm{ext}}-1} \frac{Q}{t} \cdot h\left(\left\lfloor \frac{t}{q} \cdot i \right\rceil\right) \cdot X^i$, and $(\hat{b}, \hat{\mathbf{a}}) = \frac{2N_{\mathrm{ext}}}{q} \cdot \mathrm{ct}$
for $j = 0; j < \eta; j = j + 1$ **do**

$$\mathrm{ACC}_{0,j} \leftarrow (\tau_j(T(X) \cdot X^{-\hat{b}})) \odot_N \mathbf{evk}'$$

end for
for $i = 0; i < n-1; i = i + 1$ **do**
for $j = 0; j < \eta; j = j + 1$ **do**

$$\mathrm{ACC}_{i+1,j} \leftarrow \mathrm{ACC}_{i,j} + (\mathrm{ACC}_{i,[j-\hat{a}_i]_\eta} \cdot X^{\left\lceil \frac{\hat{a}_i - j}{\eta} \right\rceil} - \mathrm{ACC}_{i,j}) \odot_N \mathbf{evk}_i$$

end for
end for

$$\mathrm{ACC}_{n,0} = \mathrm{ACC}_{n-1,0} + (\mathrm{ACC}_{n-1,[-\hat{a}_{n-1}]_\eta} \cdot X^{\left\lceil \frac{\hat{a}_{n-1}}{\eta} \right\rceil} - \mathrm{ACC}_{n-1,0}) \odot_N \mathbf{evk}_{n-1}$$

$\mathrm{ct}' \leftarrow \mathrm{SampleExtract}(\mathrm{ACC}_{n,0})$
$\mathrm{ct}' \leftarrow \mathrm{ModSwitch}_{Q\to Q_{ks}}(\mathrm{ct}')$
$\mathrm{ct}' \leftarrow \mathrm{LWE.KeySwitch}(\mathrm{ct}')$
$\mathrm{ct}' \leftarrow \mathrm{ModSwitch}_{Q_{ks}\to q}(\mathrm{ct}')$
return ct'

Proof. Let $m_0 = T(X) \cdot X^{-\hat{b}}$, $m_{0,j} = \tau_j(m_0)$; $m_i = T(X) \cdot X^{-\hat{b}+\sum_{w=0}^{i-1} \hat{a}_w \cdot s_w}$, $m_{i,j} = \tau_j(m_i)$ for $1 \leqslant i \leqslant n$. Suppose $\mathbf{evk}_i = \mathbf{g}_{e,i} \cdot f^{-1} + \mathbf{g} \cdot s_i$, $\mathrm{ACC}_{i,j} = (g_{e,i,j} + m_{i,j}) \cdot f^{-1}$.

After the first itration, we get

$$\mathrm{ACC}_{0,j} \in \mathrm{NTRU}^N_{f,Q}(\tau_j(T(X) \cdot X^{-\hat{b}})), \quad \mathrm{Var}(\mathrm{ACC}_{0,j}) \leqslant d \cdot N \cdot \frac{B^2}{12} \cdot \sigma^2.$$

By Lemma 7, we have

$$m_{i,j} \cdot X^{\left\lceil \frac{\hat{a}_i - j}{\eta} \right\rceil} = \tau_j(X^{\hat{a}_i} \cdot m_i).$$

For $0 \leqslant i \leqslant n-1$, let $\langle \mathbf{g}^{-1}(\mathrm{ACC}_{i,[j-\hat{a}_i]_\eta} \cdot X^{\left\lceil \frac{\hat{a}_i - j}{\eta} \right\rceil} - \mathrm{ACC}_{i,j}), \mathbf{g}_{e,i} \rangle = \bar{g}_{e,i,j}$, then

$$\begin{aligned}
\mathrm{ACC}_{i+1,j} &= (g_{e,i,j} + m_{i,j}) \cdot f^{-1} + s_i \cdot ((X^{\left\lceil \frac{\hat{a}_i - j}{\eta} \right\rceil} \cdot g_{e,i,[j-\hat{a}_i]_\eta} + m_{i,j} \cdot X^{\hat{a}_i}) \cdot f^{-1} \\
&\quad - (g_{e,i,j} + m_{i,j}) \cdot f^{-1}) + \langle \mathbf{g}^{-1}(\mathrm{ACC}_{i,[j-\hat{a}_i]_\eta} \cdot X^{\left\lceil \frac{\hat{a}_i - j}{\eta} \right\rceil} - \mathrm{ACC}_{i,j}), \mathbf{g}_{e,i} \rangle \cdot f^{-1} \\
&= \begin{cases} (g_{e,i,j} + \bar{g}_{e,i,j} + m_{i,j}) \cdot f^{-1} & \text{if } s_i = 0, \\ (X^{\left\lceil \frac{\hat{a}_i - j}{\eta} \right\rceil} \cdot g_{e,i,[j-\hat{a}_i]_\eta} + \bar{g}_{e,i,j} + \tau_j(m_i \cdot X^{\hat{a}_i})) \cdot f^{-1} & \text{if } s_i = 1. \end{cases}
\end{aligned}$$

Thus we have $\mathrm{ACC}_{i+1,j} \in \mathrm{NTRU}^N_{f,Q}(m_{i+1,j})$. Since the noise $\bar{g}_{e,i,j}$ satisfies

$$\mathrm{Var}(\bar{g}_{e,i,j}) \leqslant d \cdot N \cdot \frac{B^2}{12} \cdot \sigma^2,$$

clearly we can get

$$\max_{j\in[\eta]} \mathrm{Var}(\mathrm{ACC}_{i+1,j}) \leqslant \max_{j\in[\eta]} \mathrm{Var}(\mathrm{ACC}_{i,j}) + d \cdot N \cdot \frac{B^2}{12} \cdot \sigma^2.$$

After $n-1$ iterations, we have $\mathrm{ACC}_{n-1,j} \in \mathrm{NTRU}^N_{f,Q}(m_{n-1,j})$ and

$$\max_{j\in[\eta]} \mathrm{Var}(\mathrm{ACC}_{n-1,j}) \leqslant \max_{j\in[\eta]} \mathrm{Var}(\mathrm{ACC}_{0,j}) + (n-1)\cdot d\cdot N\cdot \frac{B^2}{12}\cdot\sigma^2.$$

The last Cmux gate outputs $\mathrm{ACC}_{n,0} \in \mathrm{NTRU}^N_{f,Q}(m_{n,0})$ with noise variance

$$\mathrm{Var}(\mathrm{ACC}_{n,0}) \leqslant (n+1)\cdot d\cdot N\cdot \frac{B^2}{12}\cdot\sigma^2.$$

Thus by Lemma 2 of Sample Extract, Lemma 3 of Key Switch, and Lemma 4 of ModSwitch, we get

$$\mathrm{ct}' \in \mathrm{LWE}^n_{\boldsymbol{s},q}(h(m)) \text{ and } \mathrm{Var}(\mathrm{ct}') \leqslant \frac{q^2}{Q_{ks}^2}\cdot\left(\frac{Q_{ks}^2}{Q^2}\cdot\sigma^2_{\mathrm{BR}} + \sigma^2_{\mathrm{MS1}} + \sigma^2_{\mathrm{KS}}\right) + \sigma^2_{\mathrm{MS2}}.$$

□

3.3 Sorting-based Techniques

Although our extended technique reduces the dimension of each external product from $\eta\cdot N$ to N, the total number of external-product operations remains large. Suppose that we have computed

$$\{\mathrm{ACC}_{i^*,j} \in \mathrm{NTRU}^N_{f,Q}(\tau_j(T(X)\cdot X^{-\hat{b}+\sum_{i=0}^{i^*-1}\hat{a}_i\cdot s_i}))\}_{j\in[\eta]}$$

and for $i^* \leqslant i \leqslant n-1$, we have $2^k \mid \hat{a}_i$. Recalling the final ciphertext

$$\mathrm{ACC}_{n,0} = \mathrm{ACC}_{n-1,0} + (\mathrm{ACC}_{n-1,[-\hat{a}_{n-1}]_\eta}\cdot X^{\left\lceil\frac{\hat{a}_{n-1}}{\eta}\right\rceil} - \mathrm{ACC}_{n-1,0})\odot_N \mathbf{evk}_{n-1},$$

we observe that $2^k \mid [-\hat{a}_{n-1}]_\eta$. Therefore, to correctly compute $\mathrm{ACC}_{n,0}$, it suffices to retain only the following $\eta/2^k$ ciphertexts:

$$\{\mathrm{ACC}_{n-1,j}\}_{j\in[\eta],2^k\mid j}.$$

We also recall the relation

$$\mathrm{ACC}_{i+1,j} = \mathrm{ACC}_{i,j} + (\mathrm{ACC}_{i,[j-\hat{a}_i]_\eta}\cdot X^{\left\lceil\frac{\hat{a}_i-j}{\eta}\right\rceil} - \mathrm{ACC}_{i,j})\odot_N \mathbf{evk}_i.$$

For $i^* \leqslant i \leqslant n-1$, suppose that we wish to compute $\{\mathrm{ACC}_{i+1,j}\}_{j\in[\eta],2^k\mid j}$, only $\eta/2^k$ ciphertexts

$$\{\mathrm{ACC}_{i,j}\}_{j\in[\eta],2^k\mid j}$$

are required since $2^k \mid [j-\hat{a}_i]_\eta$. Consequently, in each iteration from the i^*-th iteration through the $(n-1)$-th iteration, only $\eta/2^k$ Cmux gates need to be evaluated.rather than the η Cmux gates required by Algorithm 1.

For the general case, we can homomorphically multiply $(n-1)$ monomials $\{X^{\hat{a}_i\cdot s_i}\}_{i\in[n]}$ in any order. Let

$$I_k = \left\{i\in[n] \mid \hat{a}_i \equiv 0 \pmod{2^k}, \hat{a}_i \not\equiv 0 \pmod{2^{k+1}}\right\},\quad 0\leqslant k\leqslant \log(\eta).$$

We rewrite the sum $\sum_{i=0}^{n-1}\hat{a}_i\cdot s_i$ as

$$\sum_{k=0}^{\log\eta}\sum_{i\in I_k}\hat{a}_i\cdot s_i.$$

For $i\in I_k$, we need only to compute $\eta/2^k$ Cmux gates. The resulting bootstrapping procedure incorporating the sorting-based technique is presented in Algorithm 2.

Algorithm 2 NTRU-based Bootstrapping with Sorting-based Techniques

Input: An LWE sample $\mathrm{ct} = (b, \mathbf{a}) \in \mathrm{LWE}^n_{\mathbf{s},q}(m)$.
The evaluation keys $\{\mathbf{evk}_i\}, 0 \le i \le n, \mathbf{evk}_i \in \mathrm{VNTRU}^N_{f,Q}(s_i)$.
A special evaluation key $\mathbf{evk}' = \mathrm{VNTRU}^N_{f,Q}(f^{-1})$.
An LWE key switching key $\mathbf{ksk}_{\phi(f)\to\mathbf{s}}$ described in Lemma 3.
Output: An LWE sample $\mathrm{ct}' \in \mathrm{LWE}^n_{\mathbf{s},q}(h(m))$.

1: Set $T(X) = \sum_{i=0}^{N_{\mathrm{ext}}-1} \frac{Q}{t} \cdot h\left(\left\lfloor \frac{t}{q} \cdot i \right\rceil\right) \cdot X^i$, and $\hat{\mathrm{ct}} = (\hat{\mathbf{a}}, \hat{b}) = \frac{2N_{\mathrm{ext}}}{q} \cdot \mathrm{ct}$
for $j = 0; j < \eta; j = j + 1$ **do**

$$\mathrm{ACC}_j \leftarrow (\tau_j(T(X) \cdot X^{-\hat{b}})) \odot_N \mathbf{evk}'$$

end for
for $k = 0; k < \log(\eta) + 1; k = k + 1$ **do**
  for $i = 0; i < |I_k|; i = i + 1$ **do**
    for $j = 0; j < \eta/2^k; j = j + 1$ **do**

$$\mathrm{ACC}_{j\cdot 2^k} \leftarrow \mathrm{ACC}_{j\cdot 2^k} + \left(\mathrm{ACC}_{[j\cdot 2^k - \hat{a}_i]_\eta} \cdot X^{\left\lceil \frac{\hat{a}_i - j\cdot 2^k}{\eta} \right\rceil} - \mathrm{ACC}_{j\cdot 2^k}\right) \odot_N \mathbf{evk}_i$$

    end for
  end for
end for
3: $\mathrm{ct}' \leftarrow \mathrm{SampleExtract}(\mathrm{ACC}_0)$
4: $\mathrm{ct}' \leftarrow \mathrm{ModSwitch}_{Q\to Q_{ks}}(\mathrm{ct}')$
5: $\mathrm{ct}' \leftarrow \mathrm{LWE.KeySwitch}(\mathrm{ct}')$
6: $\mathrm{ct}' \leftarrow \mathrm{ModSwitch}_{Q_{ks}\to q}(\mathrm{ct}')$
return ct'

Theorem 2 (Sorting-based Bootstrapping)**.** *Let $h : \mathbb{Z}_t \to \mathbb{Z}_t$ be an LUT, $\mathrm{ct} \in \mathrm{LWE}^n_{\boldsymbol{s},q}(m)$ be an LWE ciphertext under a secret key $\boldsymbol{s} \leftarrow \{0,1\}^n$, and $\mathbf{evk}', \{\mathbf{evk_i}\}_{i\in[n]}, \mathbf{ksk}_{\phi(f)\to\mathbf{s}}$ are described as in Algorithm 2. Then Algorithm 2 outputs an LWE ciphertext $\mathrm{ct}' \in \mathrm{LWE}^n_{\boldsymbol{s},q}(h(m))$ with reduced noise. Its noise variance satisfies*

$$\mathrm{Var}(\mathrm{ct}') \leqslant \frac{q^2}{Q_{ks}^2} \cdot \left(\frac{Q_{ks}^2}{Q^2} \cdot \sigma^2_{\mathrm{BR}} + \sigma^2_{\mathrm{MS1}} + \sigma^2_{\mathrm{KS}}\right) + \sigma^2_{\mathrm{MS2}},$$

where $\sigma^2_{\mathrm{BR}} = (n+1) \cdot d \cdot N \cdot \frac{B^2}{12} \cdot \sigma^2$, $\sigma^2_{\mathrm{MS1}} = \frac{N+1}{12}$, $\sigma^2_{\mathrm{KS}} = N \cdot d_{ks} \cdot \sigma_L^2$, $\sigma^2_{\mathrm{MS2}} = \frac{n+1}{12}$. Here, σ^2 denotes the noise variance of NTRU evaluation keys and σ_L^2 denotes the noise variance of LWE keyswitch keys.

Proof. The proof is very similar to that of Theorem 1, so we omit it here. □

Security Analysis. The security of our bootstrapping schemes primarily relies on the hardness of the decisional LWE problem (Definition 1), the decisional NTRU problem (Definition 2), and the decisional vectorized NTRU problem (Definition 3). In addition, we require a circular-security assumption, since the initialization procedure generates a vectorized NTRU ciphertext encrypting f^{-1} under the secret key f.

4 Bootstrapping for Multi-Key Homomorphic Encryption

In this section, we focus on improving the efficiency of multi-key bootstrapping. We begin by reviewing the existing NTRU-based multi-key bootstrapping scheme proposed in [XZW+25].

4.1 A brief review of NTRU-based MK-FHE

Multi-key bootstrapping consists of the same five steps as its single-key counterpart, as described in Section 2.4. Figure 2 illustrates the overall procedure. We first recall several results that play key roles in the construction.

$$\begin{array}{c}
\mathrm{M.LWE}^{n}_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}(m) \\
\downarrow \text{M.Initialization} \\
\mathrm{M.NTRU}^{N}_{\{s_i\}_{1\leqslant i\leqslant k},Q}(T(X)\cdot X^{-\frac{2N}{q}\cdot b}) \\
\downarrow \text{M.Blind Rotation} \\
\mathrm{M.NTRU}^{N}_{\{s_i\}_{1\leqslant i\leqslant k},Q}(T(X)\cdot X^{-\frac{2N}{q}\cdot (b+\sum_{i=1}^{k}\langle \mathbf{a}_i,\mathbf{z}_i\rangle)}) \\
\downarrow \text{M.SampleExtract} \\
\mathrm{M.LWE}^{N}_{\{\phi(s_i)\}_{1\leqslant i\leqslant k},Q}(h(m)) \\
\downarrow \text{M.ModSwitch+M.KeySwitch} \\
\mathrm{M.LWE}^{n}_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}(h(m))
\end{array}$$

Figure 2: Multi-key NTRU-based Functional Bootstrapping

Lemma 9 (Multi-key SampleExtract)**.** *Suppose we have a Multi-key NTRU ciphertext*

$$(c_1,...,c_k)\in \mathrm{M.NTRU}^{N}_{\{s_i\}_{1\leqslant i\leqslant k},Q}(m),\ \text{where } c_i=\sum_{j=0}^{N-1}c_{i,j}\cdot X^j, m=\sum_{j=0}^{N-1}m_j\cdot X^j.$$

The Multi-key SampleExtract Algorithm outputs a Multi-key LWE Ciphertext ct as

$$\begin{aligned}\mathrm{M.SampleExtract}(c_1,...,c_k)=&(0,c_{1,0},-c_{1,N-1},...,-c_{1,1},...,c_{k,0},-c_{k,N-1},-c_{k,1})\\ &\in \mathrm{M.LWE}^{N}_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}(m_0)\end{aligned}$$

Lemma 10 (Multi-key KeySwitch [XZW+25, KMS24])**.** *Let Q_{ks}, B_{ks} be two positive integers and $d_{ks}=\lceil \log_{B_{ks}} Q_{ks}\rceil$. Suppose we have an M.LWE ciphertext* $\mathrm{ct}=(b,\mathbf{a}_1,...,\mathbf{a}_k)\in \mathrm{M.LWE}^{k\cdot N}_{\{\phi(s_i)\}_{i=1}^{k},Q_{ks}}(m)$ *with noise variance* $\mathrm{Var}(\mathrm{ct})$*, $\phi(s_i)=(s_{i,0},...,s_{i,N-1})$, and the switching keys* $\mathbf{ksk}_{\{\phi(s_i)\}_{i=1}^{k}\to\{\mathbf{z}_i\}_{i=1}^{k}}=\{\mathrm{ksk}_{i,j,\ell,v}\in \mathrm{LWE}^{n}_{\mathbf{z}_i,Q_{ks}}\left(vs_{i,j}B_{ks}^{\ell}\right)\}$*, where $v\in[B_{ks}]$, $j\in[N]$, $\ell\in[d_{ks}]$, $1\leqslant i\leqslant k$, with noise variance* $\mathrm{Var}(\mathbf{ksk})$*. Let* $\mathbf{a}_i=(a_{i,0},...,a_{i,N-1})$*, $\mathbf{g}_{ks}^{-1}(a_{i,j})=(a_{i,j,0},...,a_{i,j,d_{ks}-1})$, where the gadget vector $\mathbf{g}_{ks}=(1,B_{ks},...,B_{ks}^{d_{ks}-1})$. First we take* $\mathrm{ksk}_{i,j,\ell,a_{i,j,\ell}}=(\hat{b}_{i,j,\ell},\hat{\mathbf{a}}_{i,j,\ell})$*, for $1\leqslant i\leqslant k$, $j\in[N]$, $\ell\in[d_{ks}]$, and compute*

$$\mathbf{a}'_i=\sum_{j=0}^{N-1}\sum_{\ell=0}^{d_{ks}-1}\hat{\mathbf{a}}_{i,j,\ell}, b'=b-\sum_{i=1}^{k}\sum_{j=0}^{N-1}\sum_{\ell=0}^{d_{ks}-1}\hat{b}_{i,j,\ell}.$$

Then the Multi-key LWE key switching algorithm outputs

$$\mathrm{M.KeySwitch}_{\{\phi(s_i)\}_{i=1}^{k}\to\{\mathbf{z}_i\}_{i=1}^{k}}(\mathrm{ct})=(b',\mathbf{a}'_1,...,\mathbf{a}'_k)\in\mathbb{Z}_q^{k\cdot n+1},$$

which satifies $\mathrm{ct}'=\mathrm{KeySwitch}_{\phi(f)\to\mathbf{s}}(\mathrm{ct})\in \mathrm{LWE}^{n}_{\mathbf{s},Q_{ks}}(m)$*. We also have the noise variance*

$$\mathrm{Var}\left(\mathrm{ct}'\right)\leqslant \mathrm{Var}(\mathrm{ct})+k\cdot N\cdot d_{ks}\cdot \mathrm{Var}(\mathbf{ksk}).$$

Remark. In our implementation (see Section 5), we use the same method as in [XZW+25] to pack each N LWE ciphertexts into one RLWE ciphertexts.

Lemma 11 (Multi-key ModSwitch [XZW+25, KMS24])**.** *Given a multi-key LWE ciphertext*

$$\mathrm{ct} = (b, \mathbf{a}_1, \cdots, \mathbf{a}_k) \in \mathrm{M.LWE}^N_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k}, Q}(m),$$

the multi-key LWE modulus switching procedure $\mathrm{M.ModSwitch}_{Q\to q}$ *is defined as*

$$\mathrm{M.ModSwitch}_{Q\to q}(\mathrm{ct}) = \left(\left\lfloor \frac{q}{Q} b \right\rceil, \left\lfloor \frac{q}{Q} \mathbf{a}_1 \right\rceil, ..., \left\lfloor \frac{q}{Q} \mathbf{a}_k \right\rceil\right).$$

Then we have

$$\mathrm{ct}' = \mathrm{M.ModSwitch}_{Q\to q}(\mathrm{ct}) \in \mathrm{M.LWE}^N_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k}, q}(m)$$

with noise variance

$$\mathrm{Var}(\mathrm{ct}') \leqslant \left(\frac{q}{Q}\right)^2 \cdot \mathrm{Var}(\mathrm{ct}') + \frac{1 + N \cdot \sum_{i=1}^k \|\mathbf{z}_i\|_\infty}{12}.$$

We also introduce an operation called Hybrid Product, which is the core operation in the step M.Blind Rotation. The detailed process is given in Appendix A.

Lemma 12 (Hybrid Product [XZW+25])**.** *Let* $\mathbf{c} = (c_1, \cdots, c_k) \in R^k_{Q,N}$ *be a Multi-key NTRU ciphertext:*

$$\mathbf{c} \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k}, Q}(m).$$

For an index $1 \leqslant i \leqslant k$ *and a message* μ*, given a uni-encryption* $(\mathbf{d}_i, \mathbf{f}_i) \in \mathrm{UniEnc}(\mu, s_i)$ *and public keys* $\{\mathbf{b}_j\}_{1\leqslant j\leqslant k}$ *(described in Appendix A), the hybrid product* $\mathrm{HbProd}(\mathbf{c}, \{\mathbf{b}_j\}_{1\leqslant j\leqslant k}, (\mathbf{d}_i, \mathbf{f}_i))$ *outputs* $\mathbf{c}' = (c'_1, c'_2, ..., c'_k)$ *which satisfies*

$$\mathbf{c}' \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k}, Q}(\mu \cdot m)$$

and $\mathrm{Var}(\mathbf{c}') \leqslant \mathrm{Var}(\mathbf{c}) + (1 + 2k \cdot N) \cdot d \cdot N \cdot \dfrac{B^2}{12} \cdot \sigma^2$*, where* σ^2 *denotes the variance of the error distribution* χ_e*.*

4.2 Multi-key Bootstrapping with Extended and Sorting-based Techniques

Before presenting our multi-key bootstrapping algorithm, we introduce the CMux gate used in the multi-key setting. The proof is similar to Lemma 8.

Lemma 13 (Multi-Key Cmux Gate)**.** *Let* a *be a positive integer. If there are* η *ciphertexts*

$$\{\mathrm{ACC}_\ell \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k}, Q}(\tau_\ell(m))\}_{j\in[\eta]},$$

uni-encryption $\mathbf{evk} \in \mathrm{UniEnc}(z, s_i)$ *for* $1 \leqslant i \leqslant k$*,* $z \in \{0, 1\}$*, and public keys* $\{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}$*. We can compute*

$$\overline{\mathrm{ACC}}_\ell = \mathrm{ACC}_\ell + \mathrm{HbProd}((X^{\hat{a}_{i,j}} \cdot \mathrm{ACC}_{[\ell - \hat{a}_{i,j}]_\eta} - \mathrm{ACC}_\ell), \{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}, \mathbf{evk}_{i,j}).$$

The output ciphertexts satisfies

$$\overline{\mathrm{ACC}}_\ell \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k}, Q}(\tau_j(m \cdot X^{a\cdot z})).$$

Now we describe our NTRU-based multi-key bootstrapping. Given the test polynomial $T(X) \in R_{Q,N_{\text{ext}}}$, we first generate η NTRU ciphertexts $\bar{c}_\ell \in \text{NTRU}^N_{s_1,Q}(\tau_\ell(T(X)\cdot X^{-\hat{b}}))$ as in Section 3.2. This means we directly get η multi-key NTRU ciphertexts

$$\text{ACC}_\ell = (\bar{c}_\ell, \mathbf{0}) \in \text{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k},Q}(T(X)\cdot X^{-\hat{b}}).$$

For $1 \leqslant i \leqslant k$ and $j \in [n]$ $((i,j) \neq (k, n-1))$, with the Cmux gate in Lemma 13 , we compute

$$\text{ACC}_\ell \leftarrow \text{ACC}_\ell + \text{HbProd}((X^{\hat{a}_{i,j}}\cdot \text{ACC}_{[\ell-\hat{a}_{i,j}]_\eta} - \text{ACC}_\ell), \{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}, \mathbf{evk}_{i,j}).$$

After the above $nk-1$ itrations, we get

$$\text{ACC}_\ell \in \text{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k},Q}(\tau_\ell(T(X)\cdot X^{-\hat{b}+\sum_{i=1}^k \langle \hat{\mathbf{a}}_i, \mathbf{z}_i\rangle - \hat{a}_{k,n-1}\cdot z_{k,n-1}}))$$

and by taking the final cmux gate, we get our desired ciphertext. The detailed algorithm is shown in Algorithm 3.

Algorithm 3 NTRU-based Multi-key Bootstrapping with Extended Techniques

Input: A multi-key LWE ciphertext $\text{ct} = (b, \mathbf{a}_1, \cdots, \mathbf{a}_k) \in \mathbb{Z}_q^{kn+1}$;

Public keys $\{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}$, Evaluation keys $\{\mathbf{evk}_{i,j} \in \text{UniEnc}(z_{i,j}, s_i)\}_{1\leqslant i\leqslant k, j\in[n]}$, A special key $\mathbf{evk}' \in \text{VNTRU}^N_{s_1,Q}(s_1^{-1})$, Keyswitch keys $\mathbf{ksk}_{\{\phi(s_i)\}_{i=1}^k \to \{\mathbf{z}_i\}_{i=1}^k}$;

The test polynomial $T(X) \in R_{Q,N_{\text{ext}}}$.

Output: A multi-key LWE ciphertext $\text{ct}' \in \mathbb{Z}_q^{kn+1}$.

$\hat{\text{ct}} = (\hat{b}, \hat{\mathbf{a}}_1, \cdots, \hat{\mathbf{a}}_k) \leftarrow \frac{2N_{\text{ext}}}{q}\cdot \text{ct}$

for $\ell = 0; \ell < \eta; \ell = \ell + 1$ **do**

$$\bar{c}_\ell \leftarrow (\tau_\ell(T(X)\cdot X^{-\hat{b}})) \odot_N \mathbf{evk}', \text{ACC}_\ell \leftarrow (\bar{c}_\ell, \mathbf{0}) \in R^k_{Q,N}$$

end for

for $i = 1$ **to** k **do**

  for $j = 0$ **to** $n-1$ **do**

    for $\ell = 0$ **to** $\eta - 1$ **do**

      if $(i,j) \neq (k, n-1)$ **then**

$$\text{ACC}_\ell \leftarrow \text{ACC}_\ell + \text{HbProd}((X^{\left\lceil \frac{\hat{a}_{i,j}-\ell}{\eta}\right\rceil}\cdot \text{ACC}_{[\ell-\hat{a}_{i,j}]_\eta} - \text{ACC}_\ell), \{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}, \mathbf{evk}_{i,j})$$

      end if

    end for

  end for

end for

$$\text{ACC}_0 \leftarrow \text{ACC}_0 + \text{HbProd}((X^{\left\lceil \frac{\hat{a}_{k,n-1}}{\eta}\right\rceil}\cdot \text{ACC}_{[-\hat{a}_{k,n-1}]_\eta} - \text{ACC}_0), \{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}, \mathbf{evk}_{k,n-1})$$

$\text{ct}' \leftarrow \text{M.SampleExtract}(\text{ACC}_0)$

$\text{ct}' \leftarrow \text{M.ModSwitch}_{Q\to Q_{ks}}(\text{ct}')$

$\text{ct}' \leftarrow \text{M.KeySwitch}_{\{\phi(s_i)\}_{i=1}^k \to \{\mathbf{z}_i\}_{i=1}^k}(\text{ct}')$

$\text{ct}' \leftarrow \text{M.ModSwitch}_{Q_{ks}\to q}(\text{ct}')$

return ct'

Theorem 3 (MK-FHE bootstrapping with Extended Techniques)**.** *Let q, Q be the modulus of Multi-key LWE ciphertexts and NTRU ciphertexts, respectively. Given a multi-key LWE ciphertext* $\mathrm{ct} = (b, \mathbf{a}_1, \cdots, \mathbf{a}_k) \in \mathrm{M.LWE}^n_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}(m)$ *, where* $m \in \{0,1\}$, $\mathbf{z}_i \in \{0,1\}^n$ *for* $1 \leqslant i \leqslant k$*, Algorithm 3 outputs a refreshed multi-key LWE ciphertext* $\mathrm{ct}' \in \mathrm{M.LWE}^n_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}(m)$ *with less noise. The noise variance of the output ciphertext satisfies*

$$\mathrm{Var}(\mathrm{ct}') \leqslant \frac{q^2}{Q_{ks}^2}\left(\frac{Q_{ks}^2}{Q^2}(\sigma^2_{MBR} + \sigma^2_{MMS1} + \sigma^2_{MKS}) + \sigma^2_{MMS2}\right),$$

where $\sigma^2_{MBR} = (k\cdot n\cdot(1+2k\cdot N)+1)\cdot d\cdot N\cdot\frac{B^2}{12}\sigma^2$ *is the variance of the increased noise for multi-key blind rotation,* $\sigma^2_{MMS1} = \frac{1+N\cdot k}{12}, \sigma^2_{MMS2} = \frac{1+n\cdot k}{12}$ *and* $\sigma^2_{MKS} = k\cdot N\cdot d_{ks}\cdot\sigma_L^2$ *are the variance of the increased noise for modulus switching and key switching for multi-key LWE ciphertexts described in Lemma 11 and Lemma 10, respectively, where* σ^2 *denotes the noise variance of NTRU evaluation keys and* σ_L^2 *denotes the noise variance of LWE keyswitch keys.*

Proof. After the first iteration, we have

$$\mathrm{ACC}_\ell \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k},Q}(\tau_\ell(T(X)\cdot X^{-\hat{b}})) \text{ with } \mathrm{Var}(\mathrm{ACC}_\ell) \leqslant d\cdot N\cdot\frac{B^2}{12}\cdot\sigma^2$$

for $\ell \in [\eta]$. Suppose we have ℓ MK-NTRU ciphertexts

$$\mathrm{ACC}_\ell \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k},Q}(\tau_\ell(m)), \ell \in [\eta]\,.$$

Similar to computations in the proof of Theorem 1, after the step

$$\mathrm{ACC}_\ell \leftarrow \mathrm{ACC}_\ell + \mathrm{HbProd}((X^{\hat{a}_{i,j}}\cdot\mathrm{ACC}_{[\ell-\hat{a}_{i,j}]_\eta} - \mathrm{ACC}_\ell), \{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}, \mathbf{evk}_{i,j})$$

for all $\ell \in [\eta]$, the output $\{\mathrm{ACC}'_\ell\}_{\ell\in[\eta]}$ satisfies

$$\mathrm{ACC}'_\ell \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k},Q}(\tau_\ell(X^{a_{i,j}}\cdot m))$$

with noise variance $\max_{\ell\in[\eta]}\mathrm{Var}(\mathrm{ACC}'_\ell) \leqslant d\cdot N\cdot\frac{B^2}{12}\cdot(1+2k\cdot N)\sigma^2 + \max_{\ell\in[\eta]}\mathrm{Var}(\mathrm{ACC}_\ell)$ by Lemma 12.

Therefore, after the process of $1 \leqslant i^* \leqslant k$, $0 \leqslant j^* \leqslant n-2$ and the last hybrid product, we have

$$\mathrm{ACC}_0 \in \mathrm{M.NTRU}^N_{\{s_i\}_{1\leqslant i\leqslant k},Q}(T(X)\cdot X^{-\hat{b}+\sum_{i=1}^k\langle\mathbf{a}_i,\mathbf{z}_i\rangle})$$

with noise variance

$$\mathrm{Var}(\mathrm{ACC}_0) \leqslant (k\cdot n\cdot(1+2k\cdot N)+1)\cdot d\cdot N\cdot\frac{B^2}{12}\sigma^2.$$

Since the SampleExtract step does no change the noise, after ModSwitch and KeySwitch, by Lemma 9, Lemma 11 and Lemma 10, we get

$$\mathrm{ct}' \in \mathrm{M.LWE}^N_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k},q}(h(m)) \text{ and } \mathrm{Var}(\mathrm{ct}') \leqslant \frac{q^2}{Q^2}\cdot\sigma^2_{MBR} + \sigma^2_{MMS} + \sigma^2_{MKS}. \quad \square$$

The sorting-based technique can also be applied to extended Multi-key Bootstrapping. Let

$$I_m = \left\{(i,j) \in [1,k]\times[n] \mid \hat{a}_{i,j} \equiv 0 \pmod{2^m}, \hat{a}_{i,j} \not\equiv 0 \pmod{2^{m+1}}\right\}, 0 \leqslant m \leqslant \log(\eta).$$

We rewrite the sum $\sum_{i=1}^k\sum_{j=0}^{n-1}\hat{a}_{i,j}\cdot s_{i,j}$ as $\sum_{m=0}^{\log\eta}\sum_{(i,j)\in I_m}\hat{a}_{i,j}\cdot s_{i,j}$. Our detailed algorithm is presented in Algorithm 4.

Algorithm 4 NTRU-based Multi-key Bootstrapping with Sorting-based Techniques

Input: A multi-key LWE ciphertext $\mathrm{ct} = (b, \mathbf{a}_1, \cdots, \mathbf{a}_k) \in \mathbb{Z}_q^{kn+1}$;
Public keys $\{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}$, Evaluation keys $\{\mathbf{evk}_{i,j} \in \mathrm{UniEnc}(z_{i,j}, s_i)\}_{1\leqslant i\leqslant k, j\in[n]}$, A special key $\mathbf{evk}' \in \mathrm{VNTRU}_{s_1,Q}^N(s_1^{-1})$, Keyswitch keys $\mathbf{ksk}_{\{\phi(s_i)\}_{i=1}^k \to \{\mathbf{z}_i\}_{i=1}^k}$;
The test polynomial $T(X) \in R_{Q,N_{\mathrm{ext}}}$.
Output: A multi-key LWE ciphertext $\mathrm{ct}' \in \mathbb{Z}_q^{kn+1}$.
$\hat{\mathrm{ct}} = (\hat{b}, \hat{\mathbf{a}}_1, \cdots, \hat{\mathbf{a}}_k) \leftarrow \frac{2N_{\mathrm{ext}}}{q} \cdot \mathrm{ct}$
for $\ell = 0; \ell < \eta; \ell = \ell + 1$ **do**

$$\bar{c}_\ell \leftarrow (\tau_\ell(T(X) \cdot X^{-\hat{b}})) \odot_N \mathbf{evk}', \mathrm{ACC}_\ell \leftarrow (\bar{c}_\ell, \mathbf{0}) \in R_{Q,N}^k$$

end for
for $m = 0$ **to** $\log(\eta)$ **do**
  for $(i, j) \in I_m$ **do**
    for $\ell = 0$ **to** $\eta/2^m - 1$ **do**

$$\mathrm{ACC}_{\ell\cdot 2^m} \leftarrow \mathrm{HbProd}((X^{\left\lceil \frac{\hat{a}_{i,j} - \ell\cdot 2^m}{\eta} \right\rceil} \cdot \mathrm{ACC}_{[\ell\cdot 2^m - \hat{a}_{i,j}]_\eta} - \mathrm{ACC}_{\ell\cdot 2^m}), \{\mathbf{pk}_i\}_{1\leqslant i\leqslant k}, \mathbf{evk}_{i,j}) + \mathrm{ACC}_{\ell\cdot 2^m}$$

    end for
  end for
end for
$\mathrm{ct}' \leftarrow \mathrm{M.SampleExtract}(\mathrm{ACC}_0)$
$\mathrm{ct}' \leftarrow \mathrm{M.ModSwitch}_{Q\to Q_{ks}}(\mathrm{ct}')$
$\mathrm{ct}' \leftarrow \mathrm{M.KeySwitch}_{\{\phi(s_i)\}_{i=1}^k \to \{\mathbf{z}_i\}_{i=1}^k}(\mathrm{ct}')$
$\mathrm{ct}' \leftarrow \mathrm{M.ModSwitch}_{Q_{ks}\to q}(\mathrm{ct}')$
return ct'

Theorem 4 (MK-FHE bootstrapping with Sorting-based Techniques)**.** *Let q, Q be the modulus of Multi-key LWE ciphertexts and NTRU ciphertexts, respectively. Given a multi-key LWE ciphertext $\mathrm{ct} = (b, \mathbf{a}_1, \cdots, \mathbf{a}_k) \in \mathrm{M.LWE}^n_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k}, q}(m)$, where $m \in \{0,1\}$, $\mathbf{z}_i \in \{0,1\}^n$ for $1 \leqslant i \leqslant k$, Algorithm 4 outputs a refreshed multi-key LWE ciphertext $\mathrm{ct}' \in \mathrm{M.LWE}^n_{\{\mathbf{z}_i\}_{1\leqslant i\leqslant k}, q}(m)$ with less noise. The noise variance of the output ciphertext satisfies*

$$\mathrm{Var}(\mathrm{ct}') \leqslant \frac{q^2}{Q_{ks}^2}\left(\frac{Q_{ks}^2}{Q^2}(\sigma_{MBR}^2 + \sigma_{MMS1}^2 + \sigma_{MKS}^2) + \sigma_{MMS2}^2\right),$$

where $\sigma_{MBR}^2 = (k\cdot n\cdot(1+2k\cdot N)+1)\cdot d\cdot N\cdot\frac{B^2}{12}\sigma^2$ is the variance of the increased noise for multi-key blind rotation, $\sigma_{MMS1}^2 = \frac{1+N\cdot k}{12}.\sigma_{MMS2}^2 = \frac{1+N\cdot k}{12}$ and $\sigma_{MKS}^2 = k\cdot N\cdot d_{ks}\cdot\sigma_L^2$ are the variance of the increased noise for modulus switching and key switching for multi-key LWE ciphertexts described in Lemma 11 and Lemma 10, respectively, where σ^2 denotes the noise variance of NTRU evaluation keys and σ_L^2 denotes the noise variance of LWE keyswitch keys.

Proof. The proof is very similar to that of Theorem 3, so we omit it here. □

Security Analysis. In addition to the assumptions introduced in Section 3, our construction also relies on the decisional Hint-NTRU assumption [EEN+24]. The hardness of the Hint-NTRU problem guarantees the security of NTRU-based uni-encryption. A formal definition of this problem is provided in Appendix B.

5 Comparison, Parameters and Implementations

In this section, we first provide a theoretical comparison of different bootstrapping algorithms. We then describe the concrete parameter sets used by the corresponding schemes. Finally, we present the experimental results in terms of running time and memory consumption.

5.1 Comparison

In Table 1, we compare the computational cost of the various single-key bootstrapping algorithms and multi-key bootstrapping algorithm. Since external product and hybrid product operations constitute the dominant computational bottlenecks in single-key and multi-key bootstrapping, respectively, we quantify the computational costs of these two types of bootstrapping algorithms by the numbers of external product and hybrid product operations they require. Using Lemma 7 of [BORLT26], we derive the number of external product (hybrid product) operations required in Algorithm 2 (Algorithm 4).

Table 1: Comparison of different NTRU-based bootstrapping methods

	Method	#External Products
Single-Key	FINAL [BIP+22]	$(n+1)$ over $R_{Q,N_{\text{ext}}}$
	LLWW [LLW+24]	$(\frac{n}{2}+1)$ over $R_{Q,N_{\text{ext}}}$
	Our EBS (Alg. 1)	$(\eta \cdot n + 1)$ over $R_{Q',N}$
	Our SBS (Alg. 2)	$(\frac{2\eta^2+1}{3\eta} n + \eta)$ over $R_{Q',N}$
	Method	#Hybrid Products
Multi-Key	XZW [XZW+25]	$k \cdot n$ over $R_{Q,N_{\text{ext}}}$
	Our MK-EBS (Alg. 3)	$(k \cdot \eta \cdot n - \eta + 1)$ over $R_{Q',N}$
	Our MK-SBS (Alg. 4)	$(\frac{2\eta^2+1}{3\eta} k \cdot n + \eta)$ over $R_{Q',N}$

Since the noise of bootstrapping algorithms are related to the NTRU dimension N, to get similar decryption failure probability, we should choose $Q > Q'$. This further improves the efficiency of our bootstrapping with extended and sorting-based techniques.

5.2 Parameters Selection

We instantiate the LWE parameters using the LWE estimator[1], and the NTRU parameters using the NTRU estimator[2]. All parameters are chosen to provide at least 128 bits of security. The concrete LWE and NTRU parameter sets are summarized in Table 2.

For LWE ciphertexts, we use a uniformly random binary secret distribution and set the noise standard deviation to 3.19. For both NTRU and multi-key NTRU instances, we take into account the Dense Sublattice Discovery (DSD) attack [DvW21] and the Secret Key Recovery (SKR) attack. To mitigate the DSD attack, we require the modulus Q and dimension N to satisfy $Q < N^{2.484}$, thereby avoiding the fatigue point at $Q = N^{2.484+o(1)}$. For the SKR attack, we first use the NTRU estimator to determine the BKZ block size β, and then estimate the attack cost using the cost model $T(d,\beta) := 2^{0.292\cdot\beta+16.4+\log_2(8\cdot d)}$ (in the NTRU setting $d = 2N$), where $d = 2N$ in the NTRU setting, following prior works [Klu22, XZD+23].

[1] https://github.com/malb/lattice-estimator
[2] https://github.com/WvanWoerden/NTRUFatigue

We estimate the decryption failure probability using the same formula as in [DM15]: $P_{\text{fail}} = 1 - \text{erf}(\frac{q/2t}{2\sigma'})$, where erf denotes the error function, t is the plaintext modulus, and σ' is the final noise deviation. For P128B and P128B_E, we obtain $P_{\text{fail}} < 2^{-35}$, while MP128B and MP128B_E satisfy $P_{\text{fail}} < 2^{-25}$.

Table 2: Parameters for different schemes

Parameter Set	LWE						NTRU					
	n	q	t	σ_L	Q_{ks}	B_{ks}	N	σ	η	Q	B	k
P128B	790	2^{16}	32	3.19	2^{22}	2^5	2^{15}	$\frac{1}{\sqrt{2}}$	1	$\approx 2^{28}$	2^7	1
P128B_E	790	2^{16}	32	3.19	2^{21}	2^5	2^{11}	$\frac{1}{\sqrt{2}}$	16	$\approx 2^{26}$	2^7	1
MP128B	790	2^{14}	8	3.19	2^{20}	2^5	2^{13}	$\frac{1}{\sqrt{2}}$	1	$\approx 2^{27}$	2^7	2
MP128B_E	790	2^{14}	8	3.19	2^{20}	2^5	2^{11}	$\frac{1}{\sqrt{2}}$	4	$\approx 2^{26}$	2^7	2

5.3 Experimental Results

All experiments were conducted on the same desktop equipped with a 12th Gen Intel(R) Core(TM) i7-12700 CPU running at 2.10 GHz and 32 GB of RAM, with Ubuntu 24.04 as the operating system. We implemented the proposed algorithms using CHIFHE[3] and MKFHE[4]. The code was compiled with Clang 18.1.3. For single-key bootstrapping, we use the parameter set P128B for previous works [BIP+22, LLW+24] and P128B_E for our schemes, namely Alg. 1 and Alg. 2. For multi-key bootstrapping, we use the parameter set MP128B for the previous work [XZW+25] and MP128B_E for our schemes, namely Alg. 3 and Alg. 4. All reported timings are averaged over 5000 independent runs using a single thread.

Table 3: Experimental results for different NTRU-based Bootstrapping

	Method	Key Size (MB)	Time (s)
Single-Key	FINAL [BIP+22]	373.56	0.60
	LLWW [LLW+24]	546.38	0.41
	Our EBS (Alg. 1)	**21.72**	0.48
	Our SBS (Alg. 2)	**21.72**	**0.33**
Multi-Key	**Method**	Key Size (MB)	Time (s)
	XZW [XZW+25]	336.20	0.97
	Our MK-EBS (Alg. 3)	**82.84**	0.81
	Our MK-SBS (Alg. 4)	**82.84**	**0.57**

Table 3 reports the experimental results for different bootstrapping procedures. Our bootstrapping with sorting-based techniques is 1.82× and 1.24× faster than the schemes of [BIP+22] and [LLW+24], respectively. Moreover, our multi-key bootstrapping with sorting-based techniques is 1.70× faster than the previous work [XZW+25]. We note that, for single-key bootstrapping, the key size includes $(n+1)\cdot d\cdot N\cdot \log_2 Q$ bits evaluation keys $((\frac{3n}{2}+1)\cdot d\cdot N\cdot \log_2 Q$ bits in [LLW+24]) and $2\cdot N\cdot B_{ks}\cdot d_{ks}\cdot \log_2 Q_{ks}$ bits keyswitch keys for single-key bootstrapping. For multi-key bootstrapping, the key size consists of the evaluation keys of size $(k+2+2kn)\cdot d\cdot N\cdot \log_2 Q$ bits and the key-switching keys of size $2k\cdot N\cdot B_{ks}\cdot d_{ks}\cdot \log_2 Q_{ks}$ bits. Since our method reduces the dimension by a factor of η, the resulting key size is approximately $1/\eta$ of that in previous work.

[3] https://github.com/SKLC-FHE/CHIFHE
[4] https://github.com/SKLC-FHE/MKFHE

6 Conclusions

In this paper, we present extended and sorting-based techniques that enhance NTRU-based bootstrapping schemes by eliminating the constraint $q \mid 2N$. In the single-key setting, our approach reduces key sizes and accelerates both external-product and key-switching operations. In the multi-key setting, it similarly lowers key sizes and improves the efficiency of hybrid-product and key-switching operations.

Experimental results demonstrate that the sorting-based techniques yield substantial performance improvements in both the single-key and multi-key settings. Nevertheless, their applicability remains constrained by the security requirements of the underlying NTRU parameters. To further improve efficiency, we may leverage parallel implementations or explore alternative hard problems with more computation-friendly structure.

A Hybrid Product

- **Setup**(1^λ): Given the security parameter λ, set the polynomial degree N of NTRU ciphertexts, ciphertext modulus Q, gadget vector dimension d, secret distribution χ_f defined in Definition 6 and error distribution χ_e over R_N. Generate a random vector $\mathbf{a} \leftarrow R_{Q,N}^d$. Return the public parameter $pp' = (N_{\text{ext}}, N, Q, d, \mathbf{a}, \chi_f, \chi_e)$.
- **KeyGen**(pp'): Sample k polynomial $s_i \leftarrow \chi_f, 1 \leqslant i \leqslant k$ uniformly at random s_i^{-1} exists in $R_{Q,N}$, noise $\mathbf{e}_i \leftarrow {\chi_e}^d$. Compute and output the public key $\{\mathbf{b}_i = -\mathbf{a} \cdot s_i + \mathbf{e}_i\}_{1 \leqslant i \leqslant k}$.
- **UniEnc**(μ, s): For a message $\mu \in \mathbb{Z}$ and a secret key $s \leftarrow \chi_f$, sample $r \leftarrow \chi_s$ uniformly at random, and noise $\mathbf{e}_1, \mathbf{e}_2 \leftarrow {\chi_e}^d$. Compute and output the ciphertext
$$(\mathbf{d}, \mathbf{f}) = (r \cdot \mathbf{a} + \mu \cdot \mathbf{g} + \mathbf{e}_1, \mathbf{e}_2 \cdot s^{-1} + r \cdot \mathbf{g} \cdot s^{-1}) \in R_{Q,N}^d \times R_{Q,N}^d.$$
- **HbProd**$(\mathbf{c}, \{\mathbf{b}_j\}_{1 \leqslant j \leqslant k}, (\mathbf{d}_i, \mathbf{f}_i))$: Input a ciphertext $\mathbf{c} \in \text{M.NTRU}^N_{\{s_i\}_{1 \leqslant i \leqslant k}}(m)$, the public keys $\{\mathbf{b}_j\}_{1 \leqslant j \leqslant k}$, and a uni-encryption $(\mathbf{d}_i, \mathbf{f}_i) \in \text{UniEnc}(\mu, s_i)$, return an MK-NTRU ciphertext $\mathbf{c}' \in R_{Q,N}^k$ as follows:
  1. Compute the following inner products:
  $$u_j = \langle \mathbf{g}^{-1}(c_j), \mathbf{d}_i \rangle \quad \text{for } 1 \leqslant j \leqslant k,\ v = \sum_{j=1}^{k} \langle \mathbf{g}^{-1}(c_j), \mathbf{b}_j \rangle$$
  2. Compute c'_j for $1 \leqslant j \leqslant k$ where
  $$c'_j = \begin{cases} u_i + \langle \mathbf{g}^{-1}(v), \mathbf{f}_i \rangle & \text{if } j = i, \\ u_j & \text{otherwise} \end{cases}$$
  and output $\mathbf{c}' = (c'_1, c'_2, ..., c'_k)$.

B Hint-NTRU Problem

Definition 8 (Hint-NTRU Problem[EEN+24])**.** *The decisional* Hint-NTRU$_{N,Q,\chi_f,\chi'_e}$ *problem is to distinguish the following two distributions:*

$$\left\{ \left(\mathbf{h} = \mathbf{e}_1 \cdot s^{-1}, \mathbf{a}, \mathbf{b} = \mathbf{a} \cdot s + \mathbf{e}_2\right) \;\middle|\; s \leftarrow \chi_f,\ \mathbf{e}_1, \mathbf{e}_2 \leftarrow \chi_e'^d,\ \mathbf{a} \leftarrow U(R_{Q,N}^d) \right\},$$

$$\left\{ (\mathbf{u}, \mathbf{v}, \mathbf{w}) \;\middle|\; \mathbf{u}, \mathbf{v}, \mathbf{w} \leftarrow U(R_{Q,N}^d) \right\}.$$

where χ_f, χ'_e *is same as in Definition 6.*

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